

Embeddings

❖ Vector:

- It is an ordered collection (array) of a numbers used to represent something mathematically. Each number represents a value along a particular dimension.
- In AI, vectors are commonly used to represent data such as text, images, or audio in the form that model can process and compare.

❖ Embeddings:

- Vector representations of data like text, images, audio etc. used in generative AI to convert complex information into a format that model can process.
- Representing text, images, audio as points in a continuous vector space.
- These vectors capture semantic relationships within data, allowing AI models to understand context and generate relevant outputs.
- E.g. [2,4] is a vector in 2-dimesional space, [3,5,7] is a vector in 3-dimensional space.
- But in LLMs, the embedding vectors aren't usually two or three dimensions, they're often containing hundreds or thousands of dimensions i.e. "N-dimensional vector space". We as a human can't visualize them but computers can still represent these vectors and perform mathematical computation with them.

❖ Why Vector Embeddings:

- With Tokens and Token IDs we solve the problem of turning text into numbers. But Token IDs are arbitrary values i.e. random addresses. If "apple" and "banana" are assigned with ID 40 and 1842, then model has zero clue about there features like that one is red and other is yellow and both can we eat.
- Embeddings solve this problem by replacing single random ID with long string of numbers i.e. Vector (a list of co-ordinates).

❖ Types of Vector Embeddings/ Types of Data Represented with Embeddings:

1) Word Embeddings:

- Numeric vectors which represent individual words as vectors.
- It translates single word into vectors, and similar words are placed together.
- These can be helpful to show relationships between words, like "King" and "Queen" are related in the same way as "man" and "woman".
- Common word embedding models are Word2Vec, GloVe, FastText etc.

2) Sentence and Document Embeddings:

- It represents larger pieces of text i.e. sentence by capturing the overall meaning and context of an entire sentence or document, not just individual word.
- It plays crucial role in various NLP applications, such as sentiment analysis, question answering, information retrieval etc.
- Models like Doc2Vec, USE, SBERT, and ELMO have been trained on massive amounts of pre-trained data, such as Google News and Wikipedia.

3) Image Embeddings:

- It converts images into vectors and captures visual features like shapes, colours, and textures.
- They enable tasks like image recognition, classification, object detection and other visual recognition tasks.
- Image embeddings are created using deep learning models like Convolutional Neural Networks (CNNs) which includes models like VGG, ResNet, Inception and EfficientNet. These models are pre-trained on large-scale image datasets and can be used as powerful feature extractors.
- E.g. Image embedding might help model to recognize whether a given picture is a dog or cat.

4) Audio Embeddings:

- It converts sounds into vectors and captures features like pitch, tone, rhythm.
- They are particularly used in speech recognition, audio classification, emotion detection, and music analysis.
- These are created using Recurrent Neural Networks (RNNs), CNNs, and hybrid models that combine both.
- Common models are Wav2Vec, Whisper Encoder.

❖ How to create Vector Embedding:

- Step 1: Choose an embedding model
 - Select a pre-existing embedding model suitable for our data and task.
 - For text, we might choose Word2Vec, Glove, or BERT.
- Step 2: Prepare data
 - Clean and pre-process the data which is compatible with the chosen embedding model.
 - For text, this can include tokenization and possibly pre-processing. For images, we made need to resize and normalize the pixel values etc.
- Step 3: Load or trained the embedding model

- We can either use an already trained model, which already knows how to create vector representations, or train a new model using our prepared data so it can learn how to create those representations.
- Step 4: Generate Embeddings
 - For each data point, use the trained or loaded model to generate embeddings.
 - For example, if using word embedding model, input a word to get its corresponding vector representation.

❖ Vector Representation:

- It is a way to represent a data as a vector (collection of numerical values), so that models can process, compare, and analyse data mathematically.
- E.g. Suppose we have a text “I love dogs.” an embedding model may represent it as a vector i.e. [0.21, -0.45, 0.73, 0.18, 0.52, ...]

❖ Semantic Similarity:

- It refers to how two pieces of information like words, sentences, or phrases are related in meaning, even when they used different words or sentence structures.
- It measures how close or how different two pieces are in terms of meaning and context.
- E.g. Consider two sentences, Sentence 1: “How can I reset my password?” and Sentence 2: “I forgot my password, how can I change it?”. Here the wording is different, but both sentences have a similar meaning. Now Embedding model converts both the sentences into vectors
 Vector (Sentence 1) = [0.21, 0.73, -0.14, 0.52, ...] and
 Vector (Sentence 2) = [0.24, 0.70, -0.12, 0.49, ...]. These vectors are then compared using similarity methods like Cosine Similarity, Dot product. If similarity score is high, then model considers these two sentences are semantically similar.
- Explore the [TensorFlow Projector](#) to visualize and understand how similar data points are positioned closed together, which will help to illustrate concept of semantic similarity.

❖ Real World Example:

To understand how vector representation, dimensions, and similarity work together in an AI application, go through the following Movie Recommendation System example.

Image: [Movie Recommendation System](#)